Lab Assignment 6 Report

This assignment focuses on implementing the merge() function for Merge Sort and the partition() function for Quick Sort as described in the lab instructions.

Implementation Description

Merge Function

The merge() function combines two sorted subarrays into one sorted array. It creates two temporary arrays (left and right), copies data into them, and then compares elements from both arrays, placing the smaller element back into the original array. Remaining elements are then copied over.

Partition Function

The partition() function selects the last element as the pivot. It rearranges the array so that elements smaller than or equal to the pivot are placed before it, and elements greater than the pivot are placed after it. The pivot is then placed in its correct sorted position.

Testing Procedure

The program was compiled and executed using g++. The provided main.cpp was used without modification as required. Two arrays were tested: one for Merge Sort and one for Quick Sort.

Results

The output showed that both arrays were correctly sorted. The Merge Sort output displayed elements in ascending order after recursive merging. The Quick Sort output showed correct partitioning and sorting around pivots.

Correctness Justification

The merge() function ensures correctness by always selecting the smallest available element from the two subarrays. The partition() function guarantees that the pivot is placed correctly, with all smaller elements to the left and larger elements to the right. Recursive application of these functions results in a fully sorted array.

Conclusion

Both merge() and partition() were successfully implemented and tested. The output confirms that the sorting algorithms function correctly according to the assignment requirements.

```
(base) coltonzachary@Coltons-MacBook-Air-4 A6 % g++ main.cpp
(base) coltonzachary@Coltons-MacBook-Air-4 A6 % ./a.out

arr1:
-----------------------------------------------------------------
|  52  |  17  |  38  |  29  |   1  |  45  |  71  |   4  |  63  |
-----------------------------------------------------------------


arr1(Merge Sort):
-----------------------------------------------------------------
|   1  |   4  |  17  |  29  |  38  |  45  |  52  |  63  |  71  |
-----------------------------------------------------------------


arr2:
-----------------------------------------------------------------
|  26  |  89  |  12  |   6  |  23  |  90  |   2  |  56  |  43  |
-----------------------------------------------------------------


arr2(Quick Sort):
-----------------------------------------------------------------
|   2  |   6  |  12  |  23  |  26  |  43  |  56  |  89  |  90  |
-----------------------------------------------------------------

(base) coltonzachary@Coltons-MacBook-Air-4 A6 %
```